Palm OS Programming Recipes

Adding a Bitmap Family to a Form

This information describes the simplest way to add a bitmap image to your Palm OS application. This description is specific to Palm OS 3.5 and later, but for information on earlier Palm OS versions, see the compatibility notes below.

The easiest way to add a bitmap image to your Palm OS application does not require any changes to your code. You simply use Constructor for Palm OS to update the form resource for your application.

1. Use Constructor for Palm OS to open the form that will contain the bitmap.

2. In the Project Window, create a new Bitmap Family.

3. Open the Bitmap Family resource and set the values for the properties:

    - Note that every member of the Bitmap Family must have the same Width and Height.
    - You can set one of the colors to be transparent when the bitmap is drawn to the screen.
    - Set the Bitmap ID and click **Create** to use the Bitmap editor to create the bitmap.
    - You can add bitmaps of different depths to the bitmap family by selecting **Edit > New Family Member** (also by using Ctrl-K on Windows or Cmd-K on Macintosh). You can use up to five bitmap depths in one bitmap family, but bitmap images must be stored in increasing order of bit depth, and the bit depths must be unique. For backward compatibility, you should always have a 1-bit black and white bitmap as the first bitmap in the bitmap family.

4. Using the Catalog Window, drag and drop a Form Bitmap on the form.

5. Set the Bitmap Resource ID to the ID of the Bitmap Family.

6. Save your changes, and compile your application. When you run the application, the bitmap that matches the current bit depth of the screen will simply be drawn to the form when the system draws the form.

To hide, show, and move your bitmap object, you could use the Form API calls `FrmHideObject`, `FrmShowObject`, and `FrmSetObjectPosition`. **Compatibility Notes**

Prior to Palm OS 3.5, the steps for adding a bitmap were similar, but you would simply add a black-and-white bitmap rather than a bitmap family. For Palm OS 3.5 and later, using a bitmap family allows Palm OS to select the right bit depth that is supported for the screen hardware.

If your version of Constructor has no support for Bitmap Family resources, then you need to upgrade to the version of Constructor that comes with Palm OS 4.0 SDK. Note that when you create a bitmap family in Constructor, it will look like each bitmap has its own object ID. However,

these object IDs are for use by Constructor; your code cannot access an individual member of a bitmap family.

The following related recipes describe alternative methods for drawing bitmaps to the screen:

- Drawing a Bitmap Family Member
- Using `WinPaintBitmap` Instead of `WinDrawBitmap`
- Copying a Bitmap from an Offscreen Window